

Aviroop Pal

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About

AI Engineer with hands-on experience designing and shipping real-time, production-ready AI systems. I build end-to-end pipelines that ingest and process multimodal data (audio, vision, and sensor streams), integrate large models, and expose them via scalable, containerized backends using Python, PyTorch, TensorFlow, FastAPI, and Docker to deliver reliable, high-impact solutions.

Technical Skills

- **Programming & Development:** Python, Go (Golang), C, JavaScript, FastAPI, Django, Flask, React, Node.js, Express.js, Next.js, Streamlit, Docker, Git, GitHub Actions (CI/CD pipelines), WSL, SQLAlchemy (SQLite, etc.), Pydantic, Loguru, Postman, Thunder Client, Production-ready Software Engineering, Quickly Build
- **Backend Development:** Flask, Django, FastAPI, RESTful APIs, SQLAlchemy, SQLite, Real-time System Optimization
- **Machine Learning & AI:** Machine Learning, Deep Learning, TensorFlow, PyTorch, Keras, scikit-learn, CNN, LSTM, YOLO, BERT, Mistral, LoRA, Transformers, LangChain, Retrieval-Augmented Generation (RAG), Pickle, Multimodal Data Processing
- **Data Science & Analysis:** Data Analysis, Data Visualization, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook
- **Natural Language Processing (NLP):** NLTK, Transformers, BERT, LangChain, RAG
- **DevOps, MLOps and CI/CD:** Docker, Git, GitHub Actions, CI/CD pipelines, Pydantic, Loguru, Poetry, MLFlow, Weights and Biases
- **Computer Vision:** OpenCV, CNN, YOLO
- **Communication & Collaboration:** Effective Communication, Collaboration, Team Building
- **Languages:** English, Hindi, Bengali

Experience

TherapyU

Jan 2025 - Feb 2025

AI/ML Intern

Birac, Adamas University, West Bengal

- Developed and implemented production-ready AI/ML algorithms to enhance the functionality of a web application.
- Collaborated with UI/UX and development teams to integrate AI features seamlessly.
- Conducted data analysis, preprocessing, and model optimization for performance and scalability.
- Assisted in rapidly prototyping innovative AI-based solutions and provided technical documentation.
- Technologies: Python, Machine Learning, AI Integration, Data Analysis, Web Application Development

Proption Fintech Private Limited

Software Development Intern

- Quickly built internal company tools using Go(Golang) to streamline operations.
- Designed and implemented financial algorithms, including the Supertrend indicator for NIFTY, supporting data-driven decision-making.
- Contributed to backend and frontend development by implementing new features and optimizing system performance.
- Worked on data collection, preprocessing, and analysis to support algorithm development and performance evaluation.
- Technologies: Go (Golang), Backend Development, Frontend Development, Financial Algorithms, Data Analysis

BaseApp

AI/ML Intern

- Developed a real-time system optimized Voice-to-Voice (V2V) AI system with sub-second latency for conversational AI applications.
- Built an end-to-end pipeline integrating Speech-to-Text, Large Language Models, and Text-to-Speech.
- Implemented a WebSocket-based streaming API to process multimodal data streams (audio, text) for bidirectional audio communication and real-time response generation.
- Designed a scalable asynchronous backend architecture with Redis-based session management and Dockerized deployment.
- Ensured production readiness through authentication, DoS protection, memory optimization, and GPU acceleration.
- Technologies: Python 3.11, Faster-Whisper, LiteLLM, Ollama, Piper TTS, WebSockets, Redis, Docker, REST APIs, Real-Time Systems, Machine Learning

NeoCortex LLC

AI/ML Intern (Remote)

- Contributed to AI-powered Chrome Extensions focused on automation and productivity workflows.
- Integrated browser APIs with AI-driven backend systems and REST APIs.
- Assisted in developing scalable and secure extension architectures for production deployment.
- Participated in testing, debugging, deployment workflows, and collaborative remote development.
- Technologies: JavaScript, Chrome Extension APIs, AI Integrations, REST APIs, Git/GitHub

Projects

Real-Time Voice AI System

- Engineered a low-latency conversational Voice-to-Voice AI platform supporting real-time multimodal communication.
- Integrated STT, LLM inference, and TTS into a scalable streaming architecture.
- Built asynchronous APIs and optimized GPU inference for production deployment.
- Technologies: Python, FastAPI, Redis, Docker, WebSockets, LLMs

NeuroSleepNet (v1.0)

- Developed a memory-augmented learning framework to mitigate catastrophic forgetting in AI agents
- Implemented latent replay buffers and automatic task-boundary detection mechanisms.
- Designed LLM sidecar memory injection with monitoring dashboards for memory retention analysis.
- Technologies: PyTorch, HuggingFace, LangChain, Docker, LLM Systems

Advanced Multi-Modal Crop Health Classification System

- Built a multimodal deep learning system integrating multi-spectral imagery, temporal sensor data, and environmental signals.
- Designed hybrid architectures combining CNNs, LSTMs, and cross-modal attention mechanisms.
- Improved robustness using uncertainty estimation, Focal Loss, AdamW, and Cosine Annealing.
- Technologies: Pytorch, Computer Vision , Attention Mechanism, Time Series Analysis

Cancer Digital Twin

- Developed a patient-specific cancer progression simulation framework using clinical and genomic data.
- Implemented treatment simulation and progression prediction using survival analysis and Markov models.
- Built scalable APIs for personalized prediction workflows.
- Technologies: Python, FastAPI, Survival Analysis, Machine Learning, Clinical Data Integration

Federated Learning & Differential Privacy Research

- Researching privacy-preserving federated learning architectures for decentralized AI systems.
- Exploring differential privacy techniques while maintaining high model accuracy across distributed nodes.
- Building scalable distributed training pipelines with CI/CD integration.
- Technologies: Pytorch, Tensorflow, Federated Learning , Differential Privacy

Research Paper Implementations

- Implemented transformer architectures from *Attention Is All You Need* and ResNet models from *Deep Residual Learning for Image Recognition*.
- Evaluated models on benchmark datasets, improving summarization ROUGE scores by 10% and image classification accuracy by 7%.
- Technologies: Transformers, PyTorch, TensorFlow

Education

Adamas University

BTech, Computer Science and Engineering.

- **GPA:** 8

Certification

- **Generative AI Fundamentals:** Google
- **Prepare Data for ML APIs on Google Cloud:** DeepLearning.ai
- **Create and Manage cloud resources:** Google
- **Enterprise Design Thinking:** IBM
- **Build a Secure Google Cloud Network:** Google